Handling Class Imbalance in Machine Learning

A Practical Guide using SMOTE Technique

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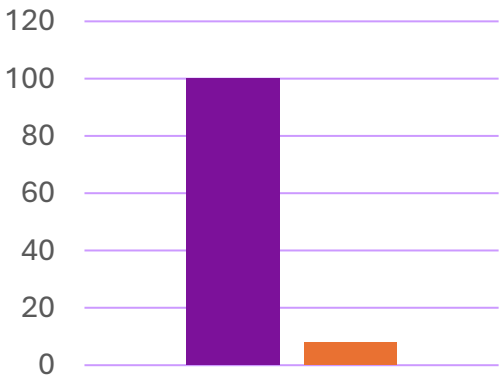
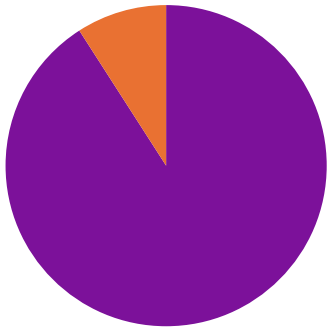
Module: Machine Learning & Neural Networks | **Tutor:** Darshan Kakkad

What is Class Imbalance?

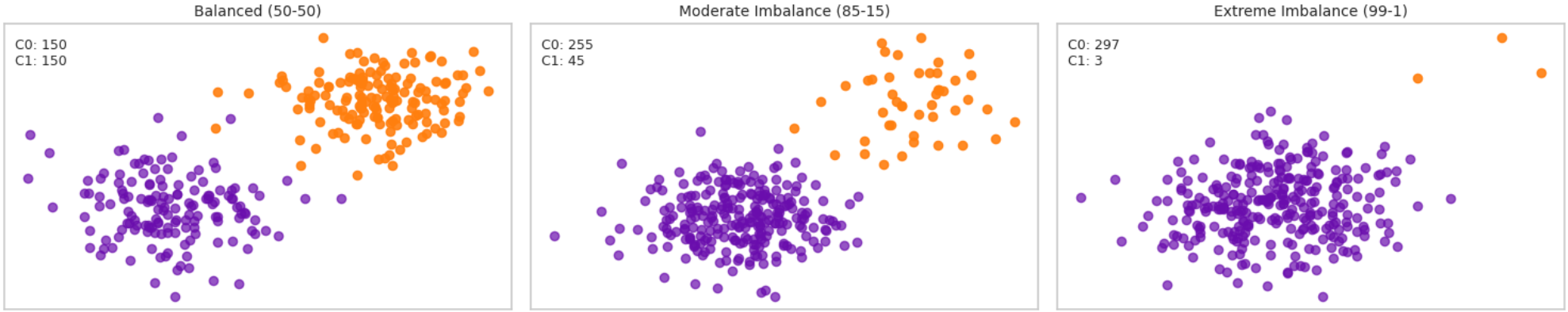
- Class imbalance occurs when one class significantly outnumbers another.
- Unequal distribution of classes
- One class dominates the dataset
- **Example:**
  - Fraud Detection: 99% non-fraud, 1% fraud
  - Spam Emails Detection





Problems of having Class Imbalance

- Model becomes biased toward majority class
- Minority class often ignored
- Poor performance in real-world use
- **Critical domains:**
  - Healthcare
  - Fraud Detection
  - Security Systems



Accuracy Problem

- Why Accuracy Fails

  Accuracy = Correct Predictions / Total Predictions

- Example:
  - 9 normal, 1 fraud
  - Predict all as normal → 90% accuracy

Case	Prediction
Authentic Transaction	Authentic Transaction
Authentic Transaction	Authentic Transaction
Authentic Transaction	Authentic Transaction
Authentic Transaction	Authentic Transaction
Authentic Transaction	Authentic Transaction
Authentic Transaction	Authentic Transaction
Authentic Transaction	Authentic Transaction
Authentic Transaction	Authentic Transaction
Fraud Transaction	Authentic Transaction

Better Metrics

- Better Evaluation Metrics
  - Precision → Correct positive predictions
  - Recall → Detecting actual positives
  - F1 Score → Balance of precision & recall

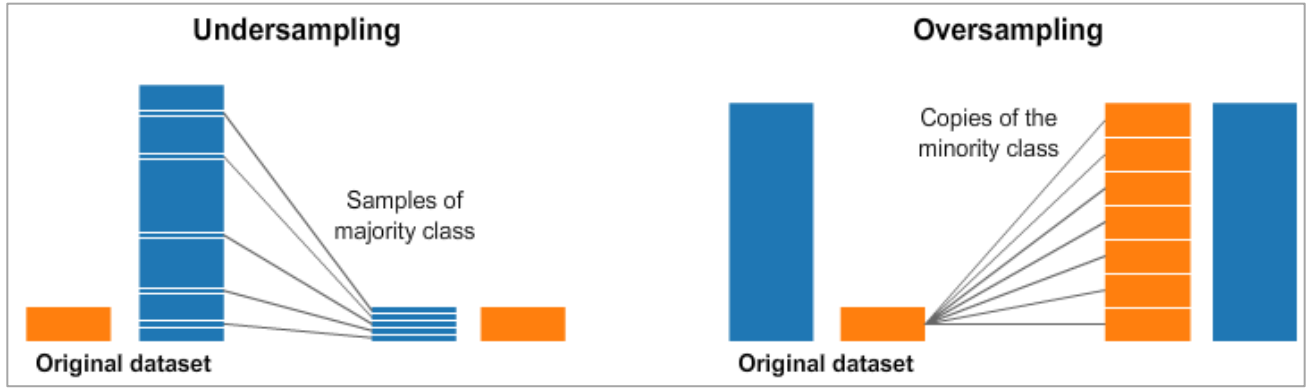
- Confusion Matrix:

$$
\begin{aligned}
precision &= \frac{TP}{TP + FP} \\
recall &= \frac{TP}{TP + FN} \\
F1 &= \frac{2 \times precision \times recall}{precision + recall} \\
accuracy &= \frac{TP + TN}{TP + FN + TN + FP} \\
specificity &= \frac{TN}{TN + FP}
\end{aligned}
$$

Overall	Actual Positive	Actual Negative
Predicted Positive	(FP)	(FP)
Predicted Negative	10	(FN)

Re-sampling Techniques

- Over-sampling:
  - Increase minority class samples
- Under-sampling:
  - Reduce majority class samples



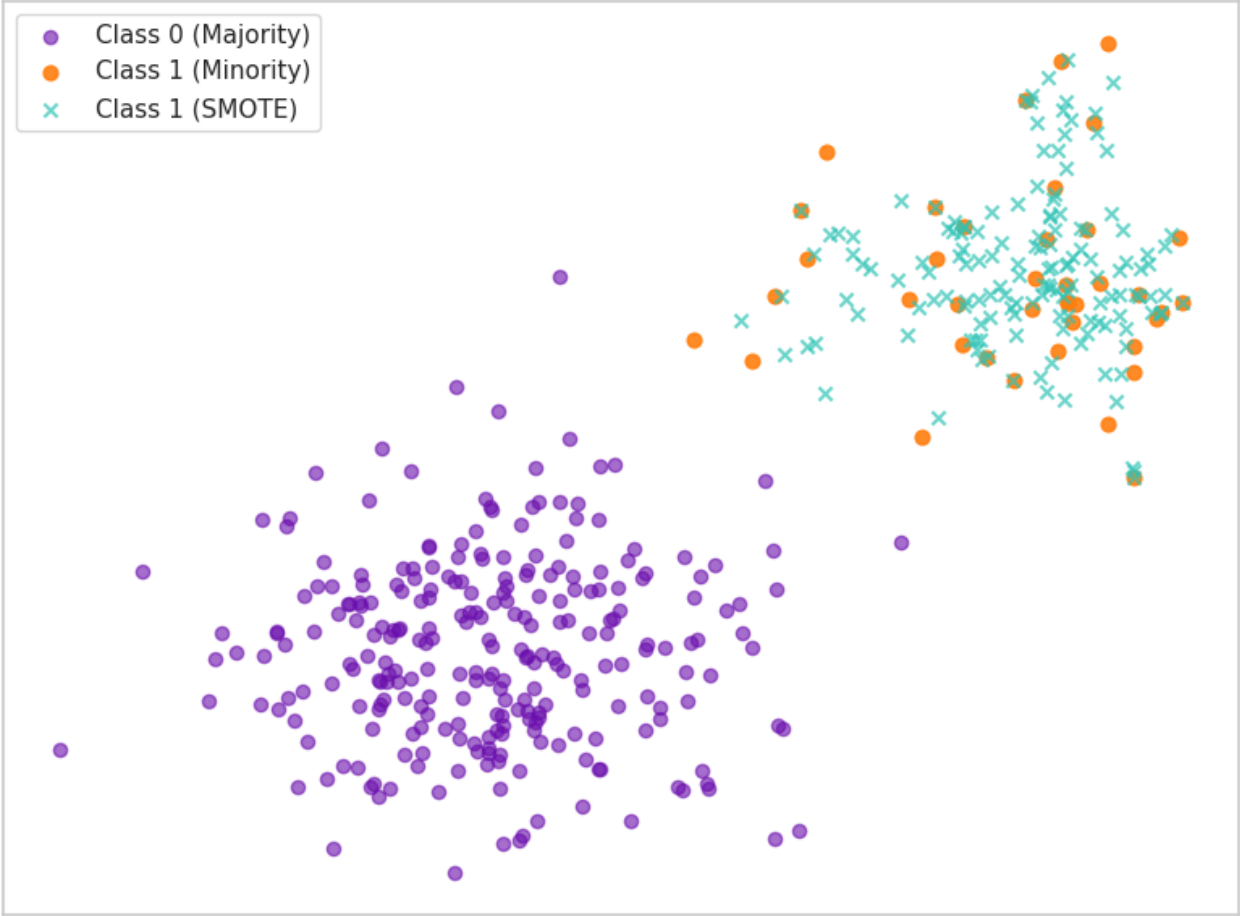
Trade-offs:

- Over-sampling → Overfitting risk
- Under-sampling → Loss of data

SMOTE

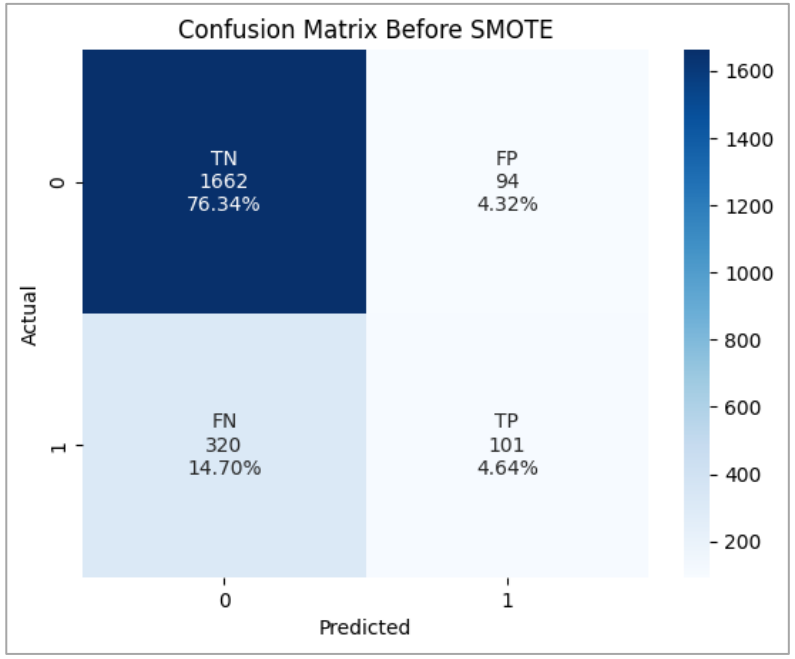
- Synthetic Minority Oversampling Technique

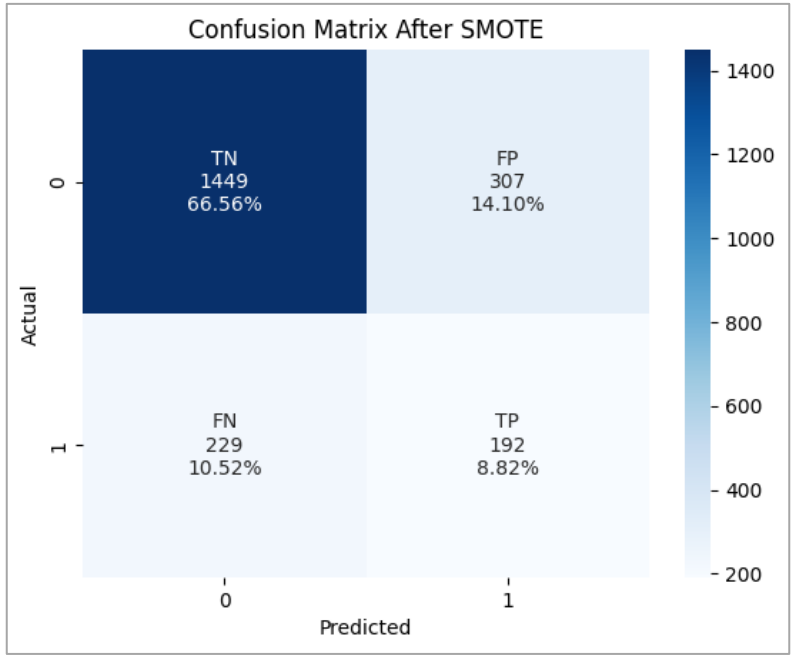
- Generates synthetic data points
- Uses nearest neighbours
- Better than simple duplication



Practical Example (Dataset from Kaggle)

- Dataset: Software Defect Prediction
- Applied: SMOTE & Random Forest

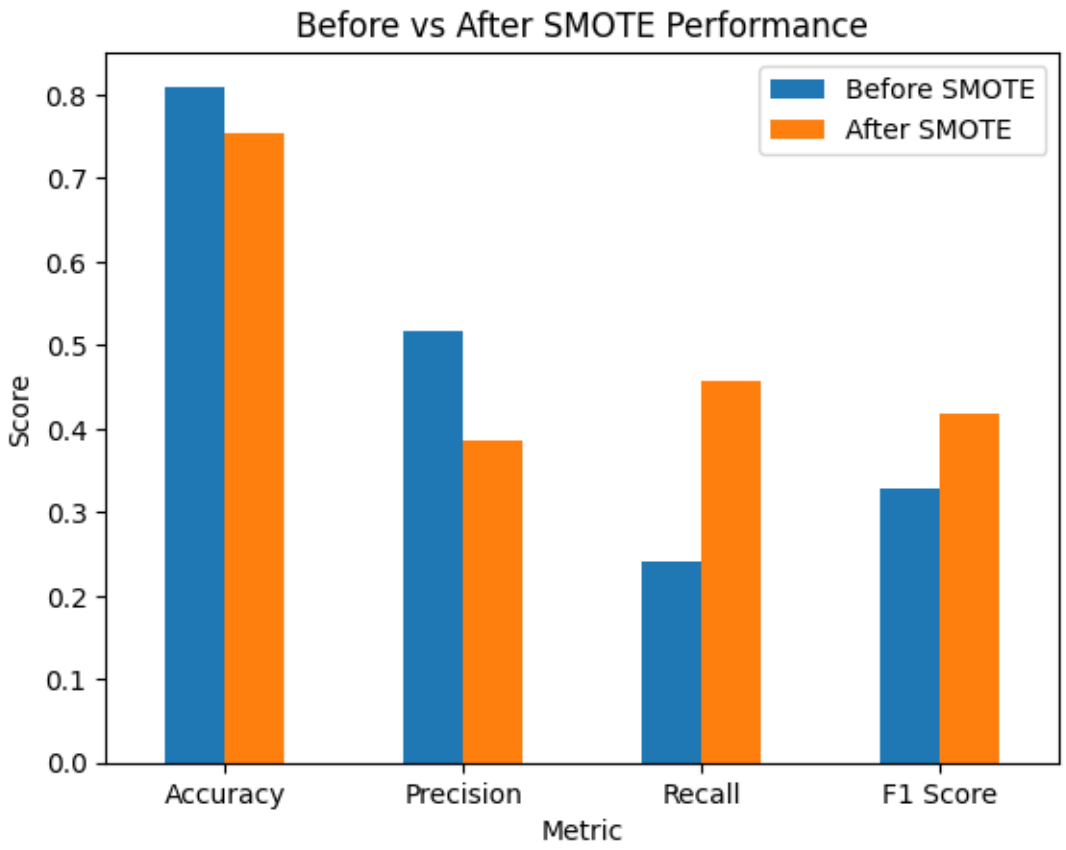




Results

- Recall improved significantly
- Better detection of minority class
- Slight trade-off in precision

Overall: More reliable model



Reference

- Kaggle Notebook (Marcin Rutecki) – For research & revise of concepts
  - Link: https://www.kaggle.com/code/marcinrutecki/best-techniques-and-metrics-for-imbalanced-dataset
- Scikit-learn Documentation – Usage of Scikit-learn properly
- Kaggle Dataset (Software Defect Prediction by Mustafa CEVIK) – To perform SMOTE Technique
  - Link: https://www.kaggle.com/datasets/semustafacevik/software-defect-prediction